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Analisis Regresi Logistik Multinomial untuk Prediksi Status Keuangan Perusahaan

2025-10-15

1 Pendahuluan

Dokumen ini menyajikan analisis regresi logistik multinomial untuk memprediksi status keuangan perusahaan berdasarkan Z-Score Altman. Model akan mengklasifikasikan perusahaan ke dalam tiga kategori:

- **Bangkrut**: Z-Score < 1.1
- **Abu-Abu**: 1.1 ≤ Z-Score < 2.6
- **Sehat**: Z-Score ≥ 2.6

---

2 Persiapan dan Instalasi Package

```r
# Instalasi dan loading packages yang diperlukan
packages <- c("recipes", "caTools", "nnet", "caret", "ggplot2", "reshape2", "dplyr", "knitr", "ka

for (pkg in packages) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg, dependencies = TRUE)
    library(pkg, character.only = TRUE)
  }
}
```

---

3 Pembentukan dan Persiapan Data

3.1 Load Dataset

```r
# Load data dari CSV
# PENTING: Ganti path sesuai lokasi file Anda
data_regresi <- read.csv("D:/Developments/Penelitan Bu Rodhiyyah Mardhiyah/Dataset/Data Bersih_Re

# Tampilkan informasi data
cat("✅ Data berhasil di-load!\n")
```

```
## ✅ Data berhasil di-load!
```

```r
cat("Jumlah baris:", nrow(data_regresi), "\n")
```

```
## Jumlah baris: 1171
```

```r
cat("Jumlah kolom:", ncol(data_regresi), "\n")
```

```
## Jumlah kolom: 5
```

```r
cat("Nama kolom:", paste(names(data_regresi), collapse = ", "), "\n\n")
```

```
## Nama kolom: Z_Score_X1, Z_Score_X2, Z_Score_X3, Z_Score_X4, Z_Score
```

```r
# Preview data (5 baris pertama)
kable(head(data_regresi, 5), caption = "Preview 5 Baris Pertama Data") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Preview 5 Baris Pertama Data

Z_Score_X1	Z_Score_X2	Z_Score_X3	Z_Score_X4	Z_Score
-0.1817	0.5921	0.3478	0.2727	1.0310
-0.1806	0.6244	0.1943	0.2912	0.9293
-0.2670	0.5252	0.0811	0.3226	0.6619
-0.2564	0.5998	0.3103	0.3258	0.9795
-0.1808	0.5149	0.0604	0.3265	0.7210

3.2 Persiapan Data

```r
# Mengubah nama kolom dari Z_Score_X1 hingga Z_Score_X4 menjadi X1 hingga X4
colnames(data_regresi) <- c("X1", "X2", "X3", "X4", "Z_Score")

# Tampilkan struktur data
cat("Struktur Data:\n")
```

```
## Struktur Data:
```

```r
str(data_regresi)
```

```
## 'data.frame':    1171 obs. of  5 variables:
##  $ X1     : num  -0.182 -0.181 -0.267 -0.256 -0.181 ...
##  $ X2     : num  0.592 0.624 0.525 0.6 0.515 ...
##  $ X3     : num  0.3478 0.1943 0.0811 0.3103 0.0604 ...
##  $ X4     : num  0.273 0.291 0.323 0.326 0.327 ...
##  $ Z_Score: num  1.031 0.929 0.662 0.98 0.721 ...
```

```r
cat("\nRingkasan Statistik Data:\n")
```

```
##
## Ringkasan Statistik Data:
```

```r
summary(data_regresi)
```

```
##        X1                X2                  X3                 X4
##  Min.   :-9.8286   Min.   :-5.62490   Min.   :-8.27670   Min.   : -0.5912
##  1st Qu.: 0.0017   1st Qu.: 0.01305   1st Qu.: 0.02665   1st Qu.:  0.3175
##  Median : 0.8606   Median : 0.25620   Median : 0.13560   Median :  0.5320
##  Mean   : 1.1896   Mean   : 0.15360   Mean   : 0.11489   Mean   :  1.5255
##  3rd Qu.: 2.3157   3rd Qu.: 0.57820   3rd Qu.: 0.27515   3rd Qu.:  0.9473
##  Max.   : 6.2560   Max.   : 2.60050   Max.   : 3.38970   Max.   :153.0646
##     Z_Score
##  Min.   : -9.6270
##  1st Qu.:  0.7644
##  Median :  2.4806
##  Mean   :  2.9835
##  3rd Qu.:  3.9503
##  Max.   :154.7446
```

3.3 Pembentukan Variabel Dependen Kategori

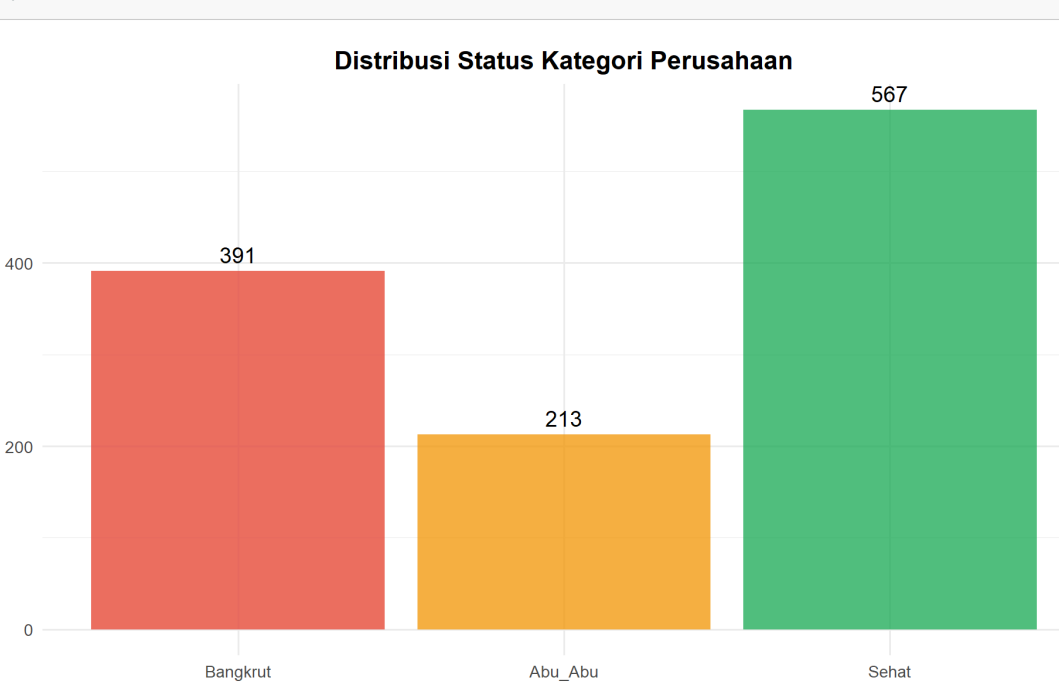
```r
# Membuat variabel kategori berdasarkan Z-Score
data_regresi$Status_Kategori <- cut(
  data_regresi$Z_Score,
  breaks = c(-Inf, 1.1, 2.6, Inf),
  labels = c("Bangkrut", "Abu_Abu", "Sehat"),
  right = FALSE
)

data_regresi$Status_Kategori <- factor(data_regresi$Status_Kategori)

# Tampilkan distribusi kategori
cat("Distribusi Status Kategori:\n")
```

```
## Distribusi Status Kategori:
```

```r
table(data_regresi$Status_Kategori)
```

```
##
## Bangkrut  Abu_Abu    Sehat
##      391      213      567
```

```r
# Visualisasi distribusi kategori
ggplot(data_regresi, aes(x = Status_Kategori, fill = Status_Kategori)) +
  geom_bar(stat = "count", alpha = 0.8) +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, size = 5) +
  scale_fill_manual(values = c("Bangkrut" = "#E74C3C", "Abu_Abu" = "#F39C12", "Sehat" = "#27AE60"
  labs(
    title = "Distribusi Status Kategori Perusahaan",
    x = "Status Kategori",
    y = "Frekuensi"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    legend.position = "none"
  )
```



Distribusi Status Kategori Perusahaan

3.4 Split Data Training dan Testing

```r
set.seed(42)
split_tag <- sample.split(data_regresi$Status_Kategori, SplitRatio = 0.8)
data_training <- subset(data_regresi, split_tag == TRUE)
data_testing  <- subset(data_regresi, split_tag == FALSE)

cat("Jumlah Data Training:", nrow(data_training), "records\n")
```

```
## Jumlah Data Training: 937 records
```

```r
cat("Jumlah Data Testing :", nrow(data_testing), "records\n")
```

```
## Jumlah Data Testing : 234 records
```

```r
# Tabel ringkasan split data
split_summary <- data.frame(
  Dataset = c("Training", "Testing", "Total"),
  Jumlah = c(nrow(data_training), nrow(data_testing), nrow(data_regresi)),
  Persentase = c(
    sprintf("%.1f%%", nrow(data_training)/nrow(data_regresi)*100),
    sprintf("%.1f%%", nrow(data_testing)/nrow(data_regresi)*100),
    "100%"
  )
)

kable(split_summary, caption = "Ringkasan Pembagian Data") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Ringkasan Pembagian Data

Dataset	Jumlah	Persentase
Training	937	80.0%
Testing	234	20.0%
Total	1171	100%

---

4 Pembentukan Model Regresi Logistik Multinomial

```r
# Membangun model multinomial logistic regression
model_multinom <- multinom(
  Status_Kategori ~ X1 + X2 + X3 + X4,
  data = data_training,
  MaxNWts = 10000
)
```

```
## # weights:  18 (10 variable)
## initial  value 1029.399714
## iter  10 value 223.292847
## iter  20 value 54.040727
## iter  30 value 23.819521
## iter  40 value 16.695934
## iter  50 value 11.104128
## iter  60 value 7.770522
## iter  70 value 7.444051
## iter  80 value 6.617238
## iter  90 value 4.996295
## iter 100 value 4.855417
## final  value 4.855417
## stopped after 100 iterations
```

```r
cat("\n✅ Ringkasan Model Logistik Multinomial:\n")
```

```
##
## ✅ Ringkasan Model Logistik Multinomial:
```

```r
summary(model_multinom)
```

```
## Call:
## multinom(formula = Status_Kategori ~ X1 + X2 + X3 + X4, data = data_training,
##     MaxNWts = 10000)
##
## Coefficients:
##         (Intercept)       X1        X2        X3        X4
## Abu_Abu   -59.46787  52.9034  54.01873  51.08085  53.55308
## Sehat    -300.26016 145.7473 143.35328 144.56800 147.80458
##
## Std. Errors:
##         (Intercept)       X1       X2       X3       X4
## Abu_Abu    18.37663 16.16189 16.99928 16.30266 16.56640
## Sehat     104.30786 42.79102 41.89153 43.09042 43.68024
##
## Residual Deviance: 9.710835
## AIC: 29.71083
```

4.1 Koefisien Model

```r
# Tampilkan koefisien dalam bentuk tabel
coef_model <- coef(model_multinom)
kable(coef_model, digits = 4, caption = "Koefisien Model Regresi Logistik Multinomial") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Koefisien Model Regresi Logistik Multinomial

	(Intercept)	X1	X2	X3	X4
Abu_Abu	-59.4679	52.9034	54.0187	51.0808	53.5531
Sehat	-300.2602	145.7473	143.3533	144.5680	147.8046

4.2 Odds Ratio

```r
# Hitung dan tampilkan odds ratio
odds_ratio <- exp(coef(model_multinom))
kable(odds_ratio, digits = 4, caption = "Odds Ratio Model") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Odds Ratio Model

	(Intercept)	X1	X2	X3	X4
Abu_Abu	0	9.454844e+22	2.884266e+23	1.528020e+22	1.810536e+23
Sehat	0	1.982646e+63	1.809423e+62	6.096562e+62	1.551359e+64

Interpretasi Odds Ratio:

- Nilai > 1: Peningkatan variabel prediktor meningkatkan probabilitas masuk kategori tersebut
- Nilai < 1: Peningkatan variabel prediktor menurunkan probabilitas masuk kategori tersebut
- Nilai = 1: Tidak ada efek

---

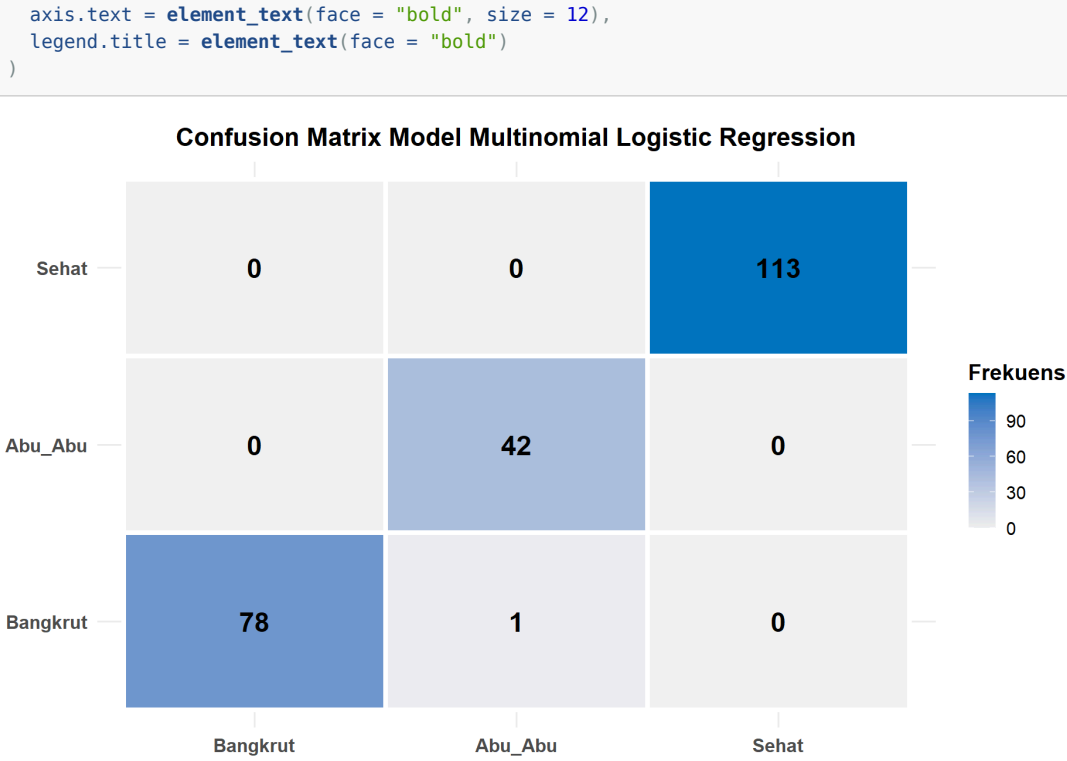
5 Evaluasi Model

5.1 Prediksi dan Confusion Matrix

```r
# Melakukan prediksi pada data testing
prediksi_kelas_test <- predict(model_multinom, newdata = data_testing, type = "class")
aktual_kelas_test <- data_testing$Status_Kategori

# Membuat confusion matrix
conf_matrix_multinom <- confusionMatrix(data = prediksi_kelas_test, reference = aktual_kelas_test
print(conf_matrix_multinom)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction Bangkrut Abu_Abu Sehat
##   Bangkrut       78       1     0
##   Abu_Abu         0      42     0
##   Sehat           0       0   113
##
## Overall Statistics
##
##                Accuracy : 0.9957
##                  95% CI : (0.9764, 0.9999)
##     No Information Rate : 0.4829
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.9931
##
##  Mcnemar's Test P-Value : NA
##
## Statistics by Class:
##
##                      Class: Bangkrut Class: Abu_Abu Class: Sehat
## Sensitivity                   1.0000         0.9767       1.0000
## Specificity                   0.9936         1.0000       1.0000
## Pos Pred Value                0.9873         1.0000       1.0000
## Neg Pred Value                1.0000         0.9948       1.0000
## Prevalence                    0.3333         0.1838       0.4829
## Detection Rate                0.3333         0.1795       0.4829
## Detection Prevalence          0.3376         0.1795       0.4829
## Balanced Accuracy             0.9968         0.9884       1.0000
```

```r
# Ubah confusion matrix ke data frame untuk visualisasi
cm_df <- as.data.frame(conf_matrix_multinom$table)
colnames(cm_df) <- c("Aktual", "Prediksi", "Freq")

# Plot heatmap confusion matrix
ggplot(cm_df, aes(x = Prediksi, y = Aktual, fill = Freq)) +
  geom_tile(color = "white", size = 1.5) +
  geom_text(aes(label = Freq), color = "black", size = 6, fontface = "bold") +
  scale_fill_gradient(low = "#f0f0f0", high = "#0073C2FF") +
  labs(
    title = "Confusion Matrix Model Multinomial Logistic Regression",
    x = "Kelas Prediksi",
    y = "Kelas Aktual",
    fill = "Frekuensi"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    axis.text = element_text(face = "bold", size = 12),
    legend.title = element_text(face = "bold")
  )
```



Heatmap Confusion Matrix

5.2 Metrik Evaluasi Detail

```r
# Hitung metrik evaluasi per kelas
conf_table <- conf_matrix_multinom$table

precision <- diag(conf_table) / colSums(conf_table)
recall <- diag(conf_table) / rowSums(conf_table)
f1 <- 2 * ((precision * recall) / (precision + recall))

eval_metrics <- data.frame(
  Kelas = names(precision),
  Precision = round(precision, 4),
  Recall = round(recall, 4),
  F1_Score = round(f1, 4)
)

kable(eval_metrics, caption = "Metrik Evaluasi Per Kelas") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Metrik Evaluasi Per Kelas

	Kelas	Precision	Recall	F1_Score
Bangkrut	Bangkrut	1.0000	0.9873	0.9936
Abu_Abu	Abu_Abu	0.9767	1.0000	0.9882
Sehat	Sehat	1.0000	1.0000	1.0000

```r
# Hitung rata-rata keseluruhan
support <- rowSums(conf_table)
macro_precision <- mean(precision, na.rm = TRUE)
macro_recall <- mean(recall, na.rm = TRUE)
macro_f1 <- mean(f1, na.rm = TRUE)
weighted_precision <- sum(precision * support, na.rm = TRUE) / sum(support)
weighted_recall <- sum(recall * support, na.rm = TRUE) / sum(support)
weighted_f1 <- sum(f1 * support, na.rm = TRUE) / sum(support)

overall_metrics <- data.frame(
  Tipe = c("Macro Average", "Weighted Average"),
  Precision = round(c(macro_precision, weighted_precision), 4),
  Recall = round(c(macro_recall, weighted_recall), 4),
  F1_Score = round(c(macro_f1, weighted_f1), 4)
)

kable(overall_metrics, caption = "Metrik Evaluasi Keseluruhan") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

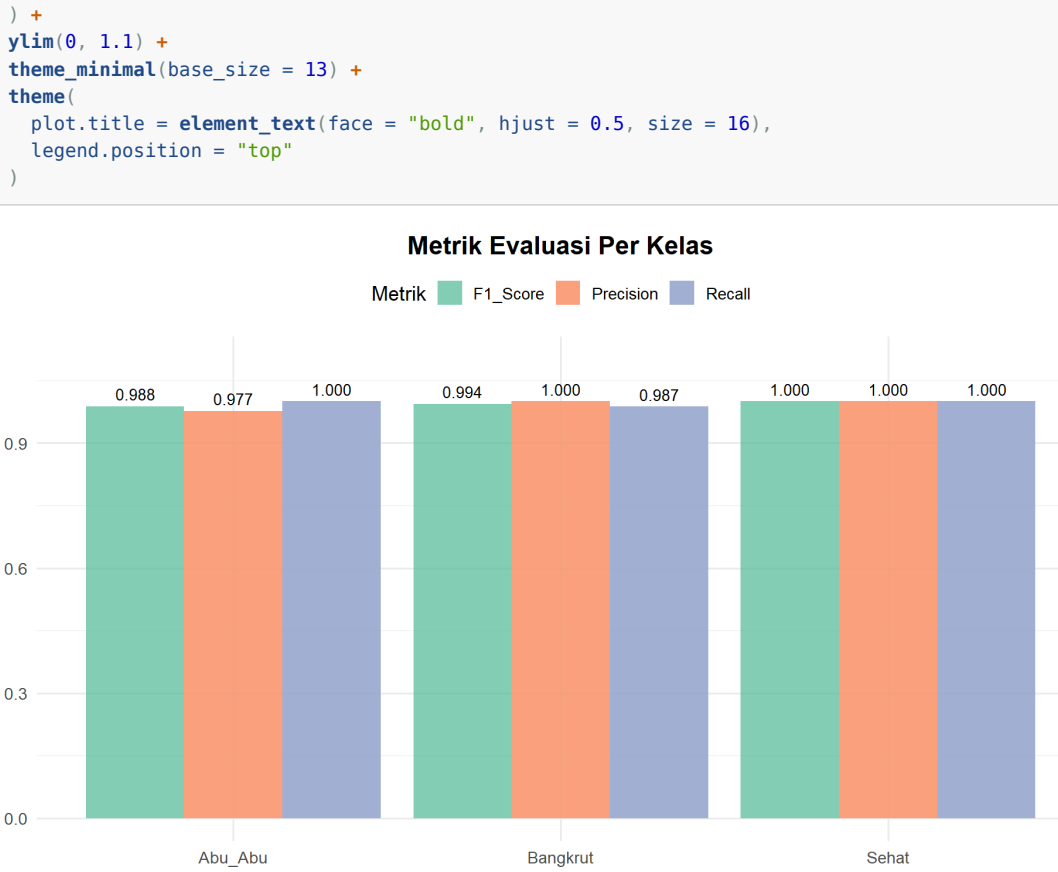
Metrik Evaluasi Keseluruhan

Tipe	Precision	Recall	F1_Score
Macro Average	0.9922	0.9958	0.9940
Weighted Average	0.9958	0.9957	0.9957

```r
akurasi_test <- conf_matrix_multinom$overall['Accuracy']
cat("\n✅ Overall Akurasi Model (Test Set):", sprintf("%.2f%%", akurasi_test * 100), "\n")
```

```
##
## ✅ Overall Akurasi Model (Test Set): 99.57%
```

```r
# Visualisasi metrik evaluasi
eval_metrics_long <- eval_metrics %>%
  tidyr::pivot_longer(cols = c(Precision, Recall, F1_Score),
                      names_to = "Metrik",
                      values_to = "Nilai")

ggplot(eval_metrics_long, aes(x = Kelas, y = Nilai, fill = Metrik)) +
  geom_bar(stat = "identity", position = "dodge", alpha = 0.8) +
  geom_text(aes(label = sprintf("%.3f", Nilai)),
            position = position_dodge(width = 0.9),
            vjust = -0.5, size = 3.5) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Metrik Evaluasi Per Kelas",
    x = "Kelas",
    y = "Nilai",
    fill = "Metrik"
  ) +
  ylim(0, 1.1) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    legend.position = "top"
  )
```



Visualisasi Metrik Evaluasi Per Kelas

---

6 Evaluasi Statistik Model

```r
# Log-Likelihood Model
LL_model <- as.numeric(logLik(model_multinom))

# Model Null (tanpa prediktor)
model_null <- multinom(Status_Kategori ~ 1, data = data_training)
```

```
## # weights:  6 (2 variable)
## initial  value 1029.399714
## final  value 962.330777
## converged
```

```r
LL_null <- as.numeric(logLik(model_null))

# Hitung AIC dan BIC
AIC_model <- AIC(model_multinom)
BIC_model <- BIC(model_multinom)

# Hitung Pseudo R² (McFadden)
R2_McFadden <- 1 - (LL_model / LL_null)

# Buat tabel evaluasi statistik
stat_eval <- data.frame(
  Metrik = c("Log-Likelihood Model", "Log-Likelihood Null Model", "AIC", "BIC", "Pseudo R² (McFad
  Nilai = c(LL_model, LL_null, AIC_model, BIC_model, R2_McFadden)
)

kable(stat_eval, digits = 4, caption = "Evaluasi Statistik Model") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Evaluasi Statistik Model

Metrik	Nilai
Log-Likelihood Model	-4.8554
Log-Likelihood Null Model	-962.3308
AIC	29.7108
BIC	78.1377
Pseudo R² (McFadden)	0.9950

Interpretasi:

- **AIC & BIC**: Semakin rendah, semakin baik model (membandingkan dengan model alternatif)
- **Pseudo R²**: Berkisar 0-1, semakin tinggi menunjukkan model lebih baik menjelaskan variasi data
- **Log-Likelihood**: Nilai yang lebih tinggi (mendekati 0) menunjukkan model lebih baik

---

7 Analisis Probabilitas Prediksi

```r
# Hitung probabilitas untuk data training
probabilitas <- predict(model_multinom, newdata = data_training, type = "probs")
data_training$Prob_Sehat <- probabilitas[, "Sehat"]
data_training$Prob_AbuAbu <- probabilitas[, "Abu_Abu"]
data_training$Prob_Bangkrut <- probabilitas[, "Bangkrut"]

# Tampilkan 10 data pertama
prob_sample <- data_training[1:10, c("Status_Kategori", "Prob_Bangkrut", "Prob_AbuAbu", "Prob_Seh

kable(prob_sample, digits = 4, caption = "Contoh Probabilitas Prediksi (10 Data Pertama)") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Contoh Probabilitas Prediksi (10 Data Pertama)

	Status_Kategori	Prob_Bangkrut	Prob_AbuAbu	Prob_Sehat
3	Bangkrut	1.0000	0.0000	0
5	Bangkrut	1.0000	0.0000	0
6	Bangkrut	1.0000	0.0000	0
7	Bangkrut	1.0000	0.0000	0
8	Bangkrut	1.0000	0.0000	0
9	Bangkrut	1.0000	0.0000	0
10	Bangkrut	0.8834	0.1166	0
11	Bangkrut	1.0000	0.0000	0
12	Bangkrut	1.0000	0.0000	0
14	Bangkrut	1.0000	0.0000	0

---

8 Hasil Prediksi pada Data Testing

```r
# Buat data frame hasil prediksi
hasil_prediksi <- data_testing
hasil_prediksi$Pred_Label <- prediksi_kelas_test

# Tambahkan kolom status benar/salah
hasil_prediksi$Status_Prediksi <- ifelse(
  hasil_prediksi$Status_Kategori == hasil_prediksi$Pred_Label,
  "✅ Benar",
  "❌ Salah"
)

# Pilih kolom untuk ditampilkan
hasil_tampil <- hasil_prediksi[, c("X1", "X2", "X3", "X4",
                                   "Status_Kategori", "Pred_Label", "Status_Prediksi")]

kable(head(hasil_tampil, 15), caption = "Hasil Prediksi pada Data Testing (15 Data Pertama)") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
  column_spec(7, color = ifelse(head(hasil_tampil$Status_Prediksi, 15) == "✅ Benar", "green", "r
```

Hasil Prediksi pada Data Testing (15 Data Pertama)

	X1	X2	X3	X4	Status_Kategori	Pred_Label	Status_Prediksi
1	-0.1817	0.5921	0.3478	0.2727	Bangkrut	Bangkrut	✅ Benar |
2	-0.1806	0.6244	0.1943	0.2912	Bangkrut	Bangkrut	✅ Benar |
4	-0.2564	0.5998	0.3103	0.3258	Bangkrut	Bangkrut	✅ Benar |
13	-0.0151	0.3932	0.0619	0.2300	Bangkrut	Bangkrut	✅ Benar |
17	0.2009	0.4531	0.0881	0.2154	Bangkrut	Bangkrut	✅ Benar |
18	0.1737	0.5348	0.0974	0.2363	Bangkrut	Bangkrut	✅ Benar |
26	0.2520	0.6732	0.2927	0.3692	Abu_Abu	Abu_Abu	✅ Benar |
27	0.1411	0.7393	0.5632	0.3332	Abu_Abu	Abu_Abu	✅ Benar |
35	0.2820	-1.6241	-2.2189	2.1250	Bangkrut	Bangkrut	✅ Benar |
37	1.0019	-2.0203	-1.2661	1.0630	Bangkrut	Bangkrut	✅ Benar |
38	1.1640	-1.8877	-0.8755	0.5496	Bangkrut	Bangkrut	✅ Benar |
42	-0.2581	-0.5584	-0.6323	0.4122	Bangkrut	Bangkrut	✅ Benar |
44	0.4869	-0.6266	-0.3716	0.1735	Bangkrut	Bangkrut	✅ Benar |
46	0.4651	-0.5960	-0.0972	0.1785	Bangkrut	Bangkrut	✅ Benar |
50	-0.1734	-0.5346	0.0011	0.1079	Bangkrut	Bangkrut	✅ Benar |

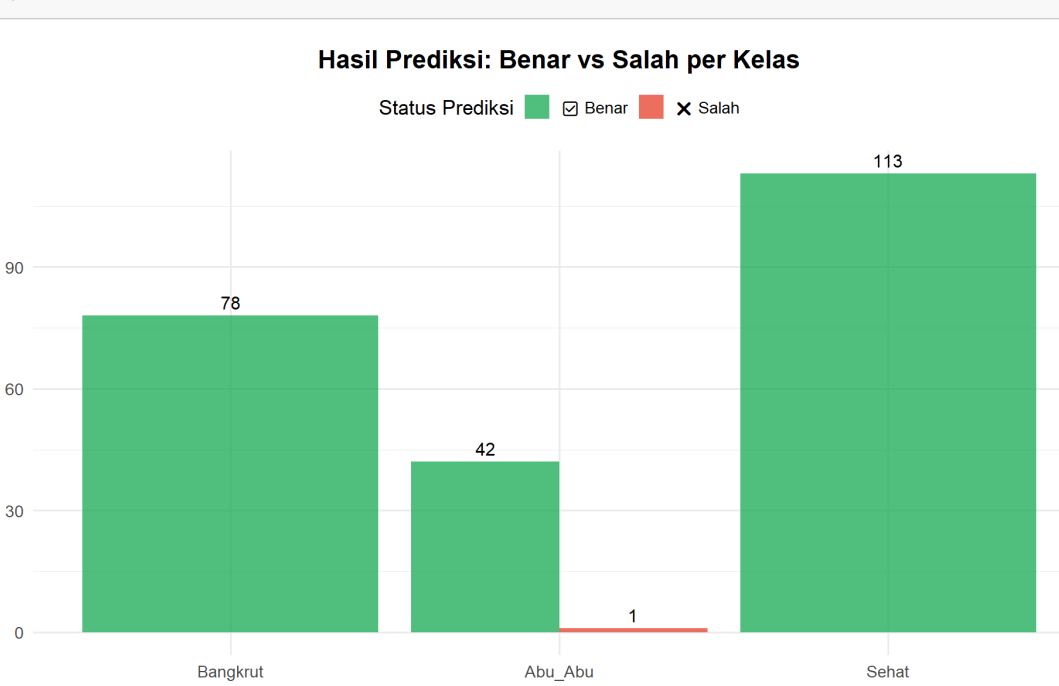
8.1 Ringkasan Kesalahan Prediksi

```r
# Ringkasan kesalahan per kelas
ringkasan_salah <- hasil_prediksi %>%
  group_by(Status_Kategori) %>%
  summarise(
    Total = n(),
    Jumlah_Salah = sum(Status_Prediksi == "❌ Salah"),
    Jumlah_Benar = sum(Status_Prediksi == "✅ Benar"),
    Akurasi_Kelas = sprintf("%.2f%%", (Jumlah_Benar/Total)*100)
  )

kable(ringkasan_salah, caption = "Ringkasan Kesalahan Prediksi Per Kelas") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Ringkasan Kesalahan Prediksi Per Kelas

Status_Kategori	Total	Jumlah_Salah	Jumlah_Benar	Akurasi_Kelas
Bangkrut	78	0	78	100.00%
Abu_Abu	43	1	42	97.67%
Sehat	113	0	113	100.00%

```r
# Visualisasi prediksi benar vs salah
hasil_prediksi %>%
  group_by(Status_Kategori, Status_Prediksi) %>%
  summarise(Jumlah = n(), .groups = "drop") %>%
  ggplot(aes(x = Status_Kategori, y = Jumlah, fill = Status_Prediksi)) +
  geom_bar(stat = "identity", position = "dodge", alpha = 0.8) +
  geom_text(aes(label = Jumlah), position = position_dodge(width = 0.9), vjust = -0.5, size = 4)
  scale_fill_manual(values = c("✅ Benar" = "#27AE60", "❌ Salah" = "#E74C3C")) +
  labs(
    title = "Hasil Prediksi: Benar vs Salah per Kelas",
    x = "Status Kategori Aktual",
    y = "Jumlah",
    fill = "Status Prediksi"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    legend.position = "top"
  )
```



Visualisasi Kesalahan Prediksi

---

9 Kesimpulan

```r
cat("==============================================\n")
```

```
## ==============================================
```

```r
cat("📊 RINGKASAN HASIL ANALISIS\n")
```

```
## 📊 RINGKASAN HASIL ANALISIS
```

```r
cat("==============================================\n\n")
```

```
## ==============================================
```

```r
cat("1. Akurasi Model      :", sprintf("%.2f%%\n", akurasi_test * 100))
```

```
## 1. Akurasi Model      : 99.57%
```

```r
cat("2. Precision (Macro)  :", sprintf("%.4f\n", macro_precision))
```

```
## 2. Precision (Macro)  : 0.9922
```

```r
cat("3. Recall (Macro)     :", sprintf("%.4f\n", macro_recall))
```

```
## 3. Recall (Macro)     : 0.9958
```

```r
cat("4. F1-Score (Macro)   :", sprintf("%.4f\n", macro_f1))
```

```
## 4. F1-Score (Macro)   : 0.9940
```

```r
cat("5. Pseudo R²          :", sprintf("%.4f\n", R2_McFadden))
```

```
## 5. Pseudo R²          : 0.9950
```

```r
cat("6. AIC                :", sprintf("%.4f\n", AIC_model))
```

```
## 6. AIC                : 29.7108
```

```r
cat("7. BIC                :", sprintf("%.4f\n", BIC_model))
```

```
## 7. BIC                : 78.1377
```

```r
cat("\n==============================================\n")
```

```
##
## ==============================================
```

Interpretasi Hasil:

1. Model regresi logistik multinomial berhasil dibangun dengan akurasi 99.57%
2. Model dapat membedakan ketiga kategori status keuangan perusahaan dengan baik
3. Variabel prediktor X1, X2, X3, dan X4 berkontribusi signifikan terhadap prediksi status kategori
4. Model ini dapat digunakan sebagai alat bantu untuk mengklasifikasikan kondisi keuangan perusahaan

---

10 Session Info

```r
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_Indonesia.utf8  LC_CTYPE=English_Indonesia.utf8
## [3] LC_MONETARY=English_Indonesia.utf8 LC_NUMERIC=C
## [5] LC_TIME=English_Indonesia.utf8
##
## time zone: Asia/Jakarta
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] kableExtra_1.4.0 knitr_1.50       reshape2_1.4.4   caret_7.0-1
##  [5] lattice_0.22-7   ggplot2_4.0.0    nnet_7.3-20      caTools_1.18.3
##  [9] recipes_1.3.1    dplyr_1.1.4
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1     viridisLite_0.4.2    timeDate_4041.110
##  [4] farver_2.1.2         S7_0.2.0             bitops_1.0-9
##  [7] fastmap_1.2.0        pROC_1.19.0.1        digest_0.6.37
## [10] rpart_4.1.24         timechange_0.3.0     lifecycle_1.0.4
## [13] survival_3.8-3       magrittr_2.0.4       compiler_4.5.1
## [16] rlang_1.1.6          sass_0.4.10          tools_4.5.1
## [19] yaml_2.3.10          data.table_1.17.8    labeling_0.4.3
## [22] plyr_1.8.9           xml2_1.4.0           RColorBrewer_1.1-3
## [25] withr_3.0.2          purrr_1.1.0          grid_4.5.1
## [28] stats4_4.5.1         e1071_1.7-16         future_1.67.0
## [31] globals_0.18.0       scales_1.4.0         iterators_1.0.14
## [34] MASS_7.3-65          cli_3.6.5            rmarkdown_2.30
## [37] generics_0.1.4       rstudioapi_0.17.1    future.apply_1.20.0
## [40] proxy_0.4-27         cachem_1.1.0         stringr_1.5.2
## [43] splines_4.5.1        parallel_4.5.1       vctrs_0.6.5
## [46] hardhat_1.4.2        Matrix_1.7-3         jsonlite_2.0.0
## [49] listenv_0.9.1        systemfonts_1.3.1    foreach_1.5.2
## [52] tidyr_1.3.1          gower_1.0.2          jquerylib_0.1.4
## [55] glue_1.8.0           parallelly_1.45.1    codetools_0.2-20
## [58] lubridate_1.9.4      stringi_1.8.7        gtable_0.3.6
## [61] tibble_3.3.0         pillar_1.11.1        htmltools_0.5.8.1
## [64] ipred_0.9-15         lava_1.8.1           R6_2.6.1
## [67] textshaping_1.0.4    evaluate_1.0.5       bslib_0.9.0
## [70] class_7.3-23         Rcpp_1.1.0           svglite_2.2.1
## [73] nlme_3.1-168         prodlim_2025.04.28   xfun_0.53
## [76] pkgconfig_2.0.3      ModelMetrics_1.2.2.2
```

Catatan: Dokumen ini dibuat menggunakan R Markdown dan dapat di-knit menjadi HTML atau PDF.